

INFO 4940/5940

HW 03 - Tune and evaluate models

HW 00 - Prefresher

HW 01 - Make a model

HW 02 - Build better data

HW 03 - Tune and evaluate models

HW 04 - Predict coffee preferences

HW 05 - Document and deploy models

HW 06 - Using LLMs to get stuff done

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HOMEWORK

MODIFIED
September 22, 2025

Important
This homework is due September 24 at 11:59pm ET.

Learning objectives

Getting started

General guidance

Attitudes towards marijuana legalization

Part 1 - Get familiar with the data

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Attitudes towards marijuana legalization

The General Social Survey is a biannual survey of the American public.

Note

The GSS gathers data on contemporary American society in order to monitor and explain trends and constants in attitudes, behaviors, and attributes. Hundreds of trends have been tracked since 1972. In addition, since the GSS adopted questions from earlier surveys, trends can be followed for up to 70 years. The GSS contains a standard core of demographic, behavioral, and attitudinal questions, plus topics of special interest. Among the topics covered are civil liberties, crime and violence, intergroup tolerance, morality, national spending priorities, psychological well-being, social mobility, and stress and traumatic events.

American permissiveness towards the legalization of marijuana has increased over time

Should be legal

Should not be legal

Source: General Social Survey Cumulative Data File

data/gss.feather contains a selection of variables from the 2022 and 2024 GSS.¹ The outcome of interest grass is a variable coded as either "should be legal" (respondent believes marijuana should be legal) or "should not be legal" (respondent believes marijuana should not be legal).

Data summary

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
colath	836	0.47	FALSE	2	yes: 473, not: 282
colmslm	841	0.47	FALSE	2	not: 546, yes: 204
degree	0	1.00	TRUE	5	hlg: 793, bac: 282, les: 207, gra: 173
fear	818	0.49	FALSE	2	no: 503, yes: 270
grass	0	1.00	FALSE	2	sho: 1105, sho: 486
gunlaw	831	0.48	FALSE	2	fav: 517, opp: 243
happy	14	0.99	TRUE	3	pre: 854, not: 389, ven: 334
health	2	1.00	FALSE	4	goo: 767, fat: 449, exc: 262, poo: 111
hispanic	15	0.99	FALSE	5	not: 1332, mex: 116, ano: 78, pue: 33
income16	179	0.89	TRUE	26	\$50: 126, \$50: 117, \$90: 109, \$17: 102
lertdie1	1211	0.24	FALSE	2	yes: 265, no: 115
owngun	812	0.49	FALSE	3	no: 507, yes: 245, ref: 27
partyid	22	0.99	TRUE	8	ind: 348, str: 261, not: 219, str: 197
polviews	99	0.94	TRUE	7	mod: 537, con: 222, sli: 199, sli: 194
pray	26	0.98	FALSE	6	sev: 587, onc: 349, nev: 238, sev: 164
pres20	538	0.66	FALSE	4	bld: 598, tru: 422, oth: 28, dld: 5
race	33	0.98	FALSE	3	whi: 1012, bla: 355, oth: 191
region	0	1.00	FALSE	9	sou: 349, eas: 288, mid: 196, pac: 189
sex	2	1.00	FALSE	2	fem: 861, mal: 728
sexfreq	409	0.74	TRUE	7	not: 411, 2 o: 168, 2 o: 157, onc: 144
wrkstat	3	1.00	FALSE	8	wor: 602, ret: 438, kee: 171, wor: 150

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
id	0	1.00	992.85	572.81	2	501	973	1492.5	1990	
age	69	0.96	52.06	18.29	18	36	53	67.0	89	
hrs1	844	0.47	40.10	15.65	0	35	40	47.0	89	
wordsum	806	0.49	5.49	2.18	0	4	6	7.0	10	

What's a feather file?

Essentially, it's a fast binary format for storing tabular data that can be read and written quickly by multiple programming languages.

R

Python

```
library(arrow)

# Result is a tibble-formatted data frame
gss <- read_feather("data/gss.feather")
```

Get detailed documentation for the variables

You can find the documentation for each of the available variables using the [GSS Data Explorer](#). Just search by the column name to find the associated description.

Part 1 - Get familiar with the data

Exercise 1

Identify appropriate performance metrics. Select an appropriate set of performance metrics for evaluating classification models for predicting grass. Justify your choices.

Note

Use these metrics throughout the rest of the assignment **except for exercises 4-5**.

Difference between J-index and balanced accuracy

The J-index is defined as sensitivity + specificity - 1, while balanced accuracy is defined as (sensitivity + specificity) / 2. They are linearly related:

$$\text{balanced accuracy} = \frac{\text{J-index} + 1}{2}$$

scikit-learn does not have a built-in metric for the J-index, but balanced accuracy is equivalent as a linear transformation. For the purposes of model evaluation and comparison, they can be used interchangeably.

Exercise 2

Import, partition, and explore the data. Import the data and reproducibly partition it into training and test sets. Briefly justify your choice of partitioning strategy. Conduct exploratory data analysis to understand the distributions of the outcome grass and some of the potential predictors. Summarize your findings.

Exercise 3

Resample the training data, fit a null model, and evaluate the results. Resample the training data using an appropriate method (justify your choice) and fit a null model to establish a baseline of comparison. Interpret the results of the null model.

Part 2 - Dealing with class imbalance

Hopefully one of the characteristics you noticed is that the majority of respondents favor legalization. This is a common issue in classification problems, known as **class imbalance** where one of the outcome classes dominates or is significantly larger than the other class(es).

One technique for handling this issue is to **downsample** the majority class. This involves randomly removing observations from the majority class until the class sizes achieve a desired ratio. But why would this be good? Isn't this just throwing away data? Didn't we learn that more is better?!!?!!

Exercise 4

Let's evaluate these claims by **comparing the performance of a model trained on the original data to one trained on a downsampled dataset**.

You will fit two **penalized regression models**. Specifically, you should estimate a Lasso regression model using all available variables (except id) to predict grass and tune over the *penalty* tuning parameter. Required data preprocessing will include:

Impute all numeric predictors with the arithmetic mean

Normalize all numeric predictors to mean of 0 and variance of 1

For the downsampled model, you will downsample the majority class to achieve a 1:1 ratio of those in favor to those opposed.

How to implement downsampling

R

Python

Use an appropriate function from the [themis](#) package to downsample the majority class.

Tune both models using 10-fold cross-validation and at least 20 possible values for the *penalty* parameter. Estimate the models' performance using sensitivity, specificity, ROC AUC, Brier class, and [J-index](#).

Looking to learn more?

Exercise 5

If our goal is to identify both the positive and negative outcomes (e.g. squirrels and no squirrels), which approach is preferred: downsampling or not? Why?

If our goal is to identify both the positive and negative outcomes (e.g. squirrels and no squirrels), which metric should we use to *finalize the model* (i.e. select the appropriate *largest* or *smallest* value)? Why?

Using metrics to select a final model

Remember that you do not have to always choose the model with the absolute largest or smallest value for a metric. Sometimes you are willing to trade-off performance for simplicity. See the [select_*\(\) functions from \(tune\)](#) for other possible approaches.³

Part 3 - tune and evaluate models

Exercise 6

Fit a random forest model. Estimate a random forest model to predict grass as a function of all the other variables in the dataset (except id). Perform any required data preprocessing and explain why it is necessary. Implement feature engineering as you wish. Tune the model across relevant hyperparameters.⁴ How does this model perform?

Exercise 7

Fit a penalized logistic regression model. Estimate a penalized logistic regression model to predict grass. Incorporate appropriate data preprocessing steps to ensure the model is estimated correctly, as well as any feature engineering steps you think are appropriate to improve the model's performance.

Tune the model across the *penalty* and *mixture* tuning parameters. If any feature engineering steps require tuning, include those as well.

Balanced grid tuning

If you are only tuning across *penalty* and *mixture*, I recommend a balanced grid tuning procedure. Because of how the *penalty* parameter is defined in a penalized regression model, you don't need to estimate separate models for each value of *penalty*. You can estimate a single model for each value of *mixture* and then extract the coefficients for all values of *penalty* from that single model. This can significantly speed up the tuning process.

Implement a balanced grid tuning procedure. Interpret the results of the tuning procedure, including how the tuning parameters effect model performance as well as the overall performance of the "best" models.

Exercise 8

Choose your own adventure. Choose and fit a third model of your choice to predict grass. This can be any model you want, but it must be different from the random forest and penalized regression models you already fit. Document your choice of model and justify why you think it is appropriate for this problem. Perform any data preprocessing or feature engineering steps you think are appropriate to improve the model's performance. Tune the model across relevant hyperparameters. How does this model perform?

Exercise 9

Pick the best performing model. Select the best performing model. Train that model using the full training set and report on it's performance using the held-out test set of data. How would you describe this model's performance at predicting attitudes towards the legalization of marijuana?

Bonus (optional) - Battle Royale

For those looking for a challenge (and a slight amount of extra credit for this assignment), train a high-performing model to predict grass.

To evaluate your model's effectiveness, you will generate predictions for a held-back **secret test set** of respondents from the survey. These can be found in `data/gss-test.rds`. The data frame has an identical structure to `gss.rds`, however **I have not included the grass column**. You will have no way of judging the effectiveness of your model, however **it has not itself**.

To evaluate your model's performance, you must create a CSV file that contains your predicted probabilities for grass. This CSV should have three columns: `id` (the id value for the respondent), `pred_should be legal`, and `pred_should not be legal`. You can generate this CSV file using the code below:

R

Python

```
augment(best_fit, new_data = gss_secret_test) |>
  select(id, starts_with("pred")) |>
  write_csv(file = "data/gss-preds.csv")
```

where `gss_secret_test` is a data frame imported from `data/gss-test.rds` and `best_fit` is the final model tested using the entire training set.

Warning

Your CSV file must

Be structured exactly as I specified above.

Be stored in the `data` folder and named `"gss-preds.csv"`.

If it does not meet these requirements, then you are not eligible to win this challenge.

The three students with the highest ROC AUC as calculated using their secret test set predictions will earn an extra (uncapped) 10 points on this homework assignment. For instance, if a student earned 45/50 points on the other components and was in the top-three, they would earn a 55/50 for this homework assignment.

Generative AI (GAI) self-reflection

As stated in the [syllabus](#), include a written reflection for this assignment of how you used GAI tools (e.g. what tools you used, how you used them to assist you with writing code), what skills you believe you acquired, and how you believe you demonstrated mastery of the learning objectives.

Render, commit, and push one last time.

Make sure that you commit and push all changed documents and your Git pane is completely empty before proceeding.

Wrap up

Submission

Go to <http://www.gradescope.com> and click *Log in* in the top right corner.

Click *School Credentials* → *Cornell University NetID* and log in using your NetID credentials.

Click on your *INFO 4940/5940 course*.

Click on the assignment, and you'll be prompted to submit it.

Mark all the pages associated with exercise. All the pages of your homework should be associated with at least one question (i.e., should be "checked").

Grading

Exercise 1: 4 points

Exercise 2: 6 points

Exercise 3: 4 points

Exercise 4: 6 points

Exercise 5: 6 points

Exercise 6: 6 points

Exercise 7: 6 points

Exercise 8: 6 points

Exercise 9: 6 points

Bonus (optional): up to 10 extra points

GAI self-reflection: 0 points

Total: 50 points

Footnotes

HW 02 - Build better data

HW 04 - Predict coffee preferences

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